Early Warning for At-Risk Students: Traditional Machine Learning versus Deep Learning on the Open University Learning Analytics Dataset

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GitHub repository: https://github.com/jagonzibwa/Intro-to-ML-Summative.git

Notebook:
https://colab.research.google.com/drive/1wsBOzl2gvDBfXgv9vYB6ZOfFHXHZhN3d?usp=sharing

Google Drive folder (dataset and saved models): https://drive.google.com/drive/folders/1ODw-Nt7TtsUmvoxONtjImycAr1a-9NgF?usp=sharing

Video presentation: https://youtu.be/aKj-4mTtk2I

Abstract

Distance-learning universities lose a large share of each cohort to failure and withdrawal, and tutors can only intervene if struggling students are identified early. This study asks whether students who will fail or withdraw can be identified using only the information a university holds 60 days after a module starts. Using the Open University Learning Analytics Dataset (32,593 enrolments and 10.6 million clickstream records), a leakage-free cohort of 26,353 enrolments still active on day 60 was built, with 54 engineered features describing student background, virtual learning environment (VLE) engagement and early assessment performance. Four traditional classifiers (logistic regression, random forest, histogram gradient boosting and a linear support vector machine) were compared with five families of neural network built in TensorFlow/Keras with `tf.data` pipelines, including a Functional-API model with categorical embeddings. On a held-out, student-disjoint test set, the best neural network reached a ROC-AUC of 0.835 and the best traditional model, an untuned logistic regression, reached 0.831; a bootstrap interval for the difference ([−0.008, 0.000]) does not exclude zero. Feature ablation showed that behaviour dominates background: demographic features alone gave a ROC-AUC of 0.649, while early assessment features alone gave 0.810. Error analysis found that most missed students looked engaged and successful at day 60, and that false-positive rates were 2.5 times higher in the most deprived areas than in the least deprived. The results suggest that, for this problem, the information available at the prediction date limits performance more than the choice of model.

Index Terms—learning analytics, early warning systems, student dropout prediction, gradient boosting, deep learning, algorithmic fairness, OULAD.

I. Introduction

Student attrition is one of the most persistent problems in higher education, and it is most severe in distance learning. Open universities report completion rates well below those of campus institutions, and many students who leave do so in the first weeks of a module [1]. Retention research links the

decision to leave to a student's academic and social integration with the institution [2]. Distance learners rarely meet tutors or peers in person, so they have fewer chances for that integration, and staff have fewer chances to notice when it is failing.

The practical response has been the early warning system: a model that scores enrolled students for risk so that tutors can direct limited support time where it is most needed. Course Signals at Purdue University [3] and the LMS analysis of Macfadyen and Dawson [4] showed that data already collected by online platforms can identify struggling students. The value of such a system depends on timing. A late prediction may be accurate but arrives after students have disengaged, while a prediction on enrolment day reaches everyone but has little behaviour to learn from.

This study builds and evaluates an early warning model for the Open University (OU), the United Kingdom's largest distance-learning university, using the Open University Learning Analytics Dataset (OULAD) [5]. As a student at African Leadership University, much of my own learning happens through online platforms, and I have seen how quickly a missed deadline or a few quiet weeks can turn into falling behind. I chose this problem because the data needed to notice that drift early already exists in most learning platforms, yet it is rarely used to prompt timely support. The research question is therefore framed around a decision a real tutor could make: Using only the information available 60 days after a module starts, can we identify which currently enrolled students will go on to fail or withdraw?

The task is formulated as binary classification, with Fail and Withdrawn forming the at-risk class. Two paradigms are compared on identical data: traditional machine learning with scikit-learn and deep learning with TensorFlow/Keras. Recent benchmarks suggest that tree ensembles still outperform neural networks on medium-sized tabular data [6], [7]; this study tests directly whether that holds for educational data with heavy-tailed engagement features.

The study makes four contributions. It constructs a leakage-free cohort and a 54-feature representation of background, engagement and early assessment, using only events up to the prediction day and splitting by student. It compares four traditional classifiers and five families of neural network under one tuning and evaluation protocol, and tests whether the best of each differs significantly. It measures what each feature family contributes and how accuracy trades against earliness. Finally, it analyses errors by outcome, module and demographic group, identifying both a limit of the data and a fairness concern.

The remainder of the report is organised as follows. Section II reviews related work on learning analytics, OULAD-based prediction and fairness. Section III describes the dataset, feature engineering and experimental design. Section IV presents the results, Section V discusses their implications and limitations, and Section VI concludes.

II. Literature Review

A. Early Warning Systems in Learning Analytics

Learning analytics applies data analysis to the traces students leave in educational systems. Macfadyen and Dawson [4] found that counts of discussion posts, emails and completed assessments in an online course explained much of the variance in final grades, and that a logistic regression on them identified most failing students. Course Signals [3] turned this kind of model into a deployed traffic-light system

combining grades, effort and prior history, and reported improved retention among warned students. Most later work follows the same template: aggregate behaviour up to a date, train on past cohorts and score current students.

Gašević et al. [8] identified an important caveat. Analysing nine courses, they found that the predictors of success varied considerably between courses with different instructional designs, and that a single model pooled across courses could misrepresent the drivers in any one course. Their finding motivates two design choices in the present study: treating the module as a feature, and reporting performance separately for each module.

B. Prediction on the OULAD Dataset

OULAD was released by the OU's Knowledge Media Institute to give researchers a realistic, openly licensed benchmark [5]. It accompanies OU Analyse, the university's own production early warning system, which combined demographic data with VLE activity and assessment results to produce weekly risk predictions for tutors [9]. Because the dataset is public, it has been used for a wide range of prediction studies.

Among traditional approaches, Hussain et al. [10] used OULAD VLE activity to predict student engagement with several classifiers and found that click-based features identified low-engagement students accurately. Among deep approaches, Hassan et al. [11] trained a long short-term memory (LSTM) network on weekly clickstream sequences to predict withdrawal and reported higher accuracy than logistic regression and a feed-forward network. Waheed et al. [12] found that deep neural networks on hand-crafted clickstream features outperformed logistic regression and support vector machines, and that accuracy grew as more weeks of data became available. Outside OULAD, Xing and Du [13] applied deep learning to MOOC dropout and argued that risk probabilities, not binary labels, should drive personalised intervention.

Three recurring weaknesses in this literature shaped the present design. Many studies use features that are known only at the end of the course, a form of data leakage [14] that inflates results and does not answer the early-warning question. Most split OULAD rows at random, so students with several enrolments can appear in both training and test sets. Comparisons also often pit a tuned neural network against untuned baselines without testing significance. This study addresses all three.

C. Deep Learning versus Tree Ensembles on Tabular Data

The wider machine learning literature suggests the margin between paradigms will be narrow. Grinsztajn et al. [6] benchmarked deep architectures against tree ensembles on 45 tabular datasets and found that tree-based models remained state of the art at around 10,000 samples, partly because neural networks are biased towards overly smooth functions and are sensitive to uninformative features. Shwartz-Ziv and Armon [7] likewise found that XGBoost generally beat recent deep tabular models. On a structured dataset of about 26,000 rows, deep learning should therefore not be expected to dominate.

One technique that is native to neural networks is the entity embedding. Guo and Berkhahn [15] showed that learning a dense vector for each level of a categorical variable can reduce memory use, speed up training and help a network generalise, particularly for high-cardinality categories. OULAD's categorical variables are few and low-cardinality, so this study tests whether embeddings help even in that unfavourable setting.

D. Fairness in Student Risk Prediction

Predictive models in education can reproduce existing inequalities. Baker and Hawn [16] reviewed evidence of algorithmic bias across educational applications for groups defined by race, gender, nationality, socio-economic status and disability, often because these attributes correlate with outcomes in historical data. Riazy et al. [17] found that at-risk models on OULAD had different error rates across demographic groups, and that removing protected attributes did not remove the disparity, because behavioural features act as proxies. This study therefore reports error rates by group as part of the evaluation.

E. Research Gap

Prior OULAD work demonstrates that both traditional and deep models can predict withdrawal and failure. Fewer studies combine a strict prediction-day cutoff, a student-disjoint split, matched tuning across paradigms, a test of statistical significance, and a subgroup error analysis in one pipeline. This study provides that combination, so that its comparison of paradigms and its account of where models fail can be relied upon.

III. Methodology

The pipeline is a single seeded Jupyter notebook that runs on a free Google Colab CPU runtime and downloads the raw data from the UCI repository at run time.

A. Dataset

OULAD [5] describes 32,593 enrolments of 28,785 anonymised students across seven modules (AAA to GGG) and four presentations in 2013 and 2014. Its seven relational tables hold demographics and final results, registration dates, 206 assessments with 173,912 submissions, and 10,655,280 daily click records over 6,364 online resources, with all dates counted from the module start. The data are real, openly licensed (CC BY 4.0) and well above the 20,000-row threshold.

The final result has four classes: Pass (37.9%), Withdrawn (31.2%), Fail (21.6%) and Distinction (9.3%). Outcomes vary strongly by module and by socio-economic background (Fig. 1). The share ending in Fail or Withdrawn ranges from about 30% in AAA to 60% in CCC and DDD, and from roughly 65% in the most deprived Index of Multiple Deprivation (IMD) band to 42% in the least deprived. Weekly VLE activity separates the outcomes from the first week (Fig. 2), so an early prediction has signal to work with.

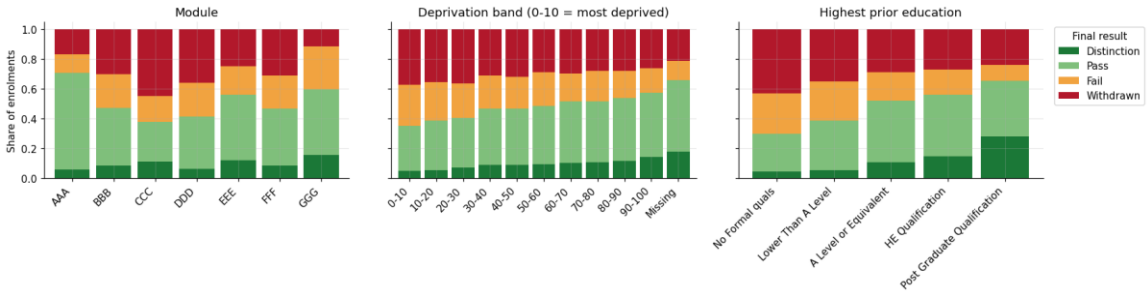


Fig. 1. Share of each final result by module (left), IMD deprivation band (centre; 0–10 = most deprived) and highest prior education (right), across all 32,593 enrolments.

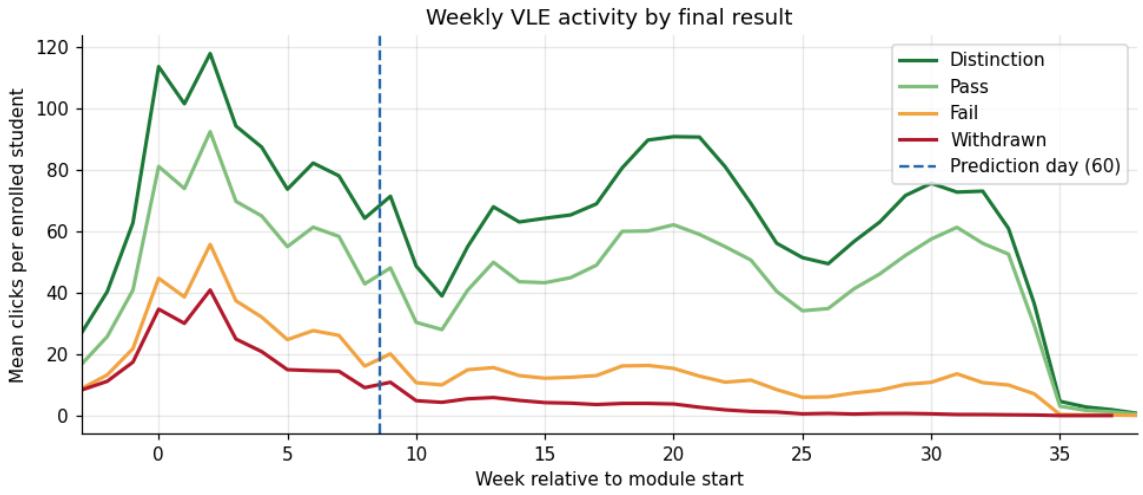


Fig. 2. Mean weekly VLE clicks per enrolled student, by final result. The dashed line marks the day-60 prediction point; only activity to its left is used as input.

B. Problem Formulation and Leakage Control

An enrolment is at risk (label 1) if its final result is Fail or Withdrawn, and on track otherwise. The merge reflects the intervention, which is the same in both cases. The cohort contains every enrolment that had registered and had not unregistered by day 60; students who have already left need no prediction and would inflate every metric with trivially easy cases. This removed 6,240 enrolments (19.1%), leaving 26,353, of which 41.6% are at risk. That is close enough to balance that no resampling was applied.

The central leakage rule [14] is that every feature uses only events dated on or before day 60. The unregistration date is used only to define the cohort and never as an input, because it encodes the Withdrawn label directly. Assessments are counted only if they were both due and submitted by the cutoff, and final exams are excluded.

C. Feature Engineering

Fifty-four features were engineered in three families (Table I). Engagement features describe not only how much a student clicks but how regularly, how recently and whether activity is fading, through a least-squares slope of weekly clicks. Clicks are also normalised by the median of the student's own module presentation, because modules demand very different amounts of online activity.

TABLE I. Engineered feature families

Family	Count	Examples
Background	13 (5 numeric, 8 categorical)	Module, semester, gender, region, highest education, IMD band, age band, disability, previous attempts, credits studied, days registered before start, concurrent modules
VLE engagement	30	Total clicks, active days, distinct resources, weekly clicks (pre-start and weeks 0–8), weekly variability

Family	Count	Examples
		and slope, days since last login, clicks by activity type, clicks relative to cohort median, no-activity flag
Assessment	11	Assessments due and submitted, submission rate, number missing, mean, minimum and weight-adjusted score, mean days submitted early, late submissions, banked results

D. Data Partitioning

The cohort was split 70/15/15 into training (18,407 enrolments), validation (3,975) and test (3,971) sets using `GroupShuffleSplit` grouped by student identifier. A student who retakes a module, or studies two at once, therefore never appears on both sides of a boundary. Assertions in the code verify this. The at-risk rate is 41.3%, 43.0% and 41.9% in the three sets. The training set is used for fitting and cross-validated search. The validation set is used for comparing models, early stopping and choosing decision thresholds. The test set was used once, after all model selection was complete.

E. Preprocessing

Preprocessing is implemented as a scikit-learn [18] `ColumnTransformer` embedded in each model pipeline. It is therefore re-fitted inside every cross-validation fold, so no statistic is computed from held-out rows. Click counts and ratios, which are heavily right-skewed (median total clicks 366, mean 583), receive median imputation, a `log1p` transform and standardisation. Other numeric features receive median imputation and standardisation. Categorical features have missing values mapped to an explicit "Unknown" level and are one-hot encoded, producing 91 model inputs.

F. Traditional Machine Learning

Four algorithm families were compared against a dummy classifier that predicts the class prior: L2-regularised logistic regression, random forest [19], histogram gradient boosting in the style of LightGBM [20], and a calibrated linear support vector machine. A kernel SVM was excluded because its training cost grows roughly quadratically with sample size. Each family was trained with default settings (experiment ML-1) and then tuned by randomised search [21] on ROC-AUC with three-fold `StratifiedGroupKFold` cross-validation, which stratifies on the target while keeping each student in one fold (ML-2; Table II).

TABLE II. Hyperparameter search spaces for traditional models

Model	Hyperparameters (range)	Candidates
Logistic regression	C (log-uniform 10^{-3} to 10^{2}); class weight (none, balanced)	12
Random forest	Trees (200, 300); max depth (none, 8, 12, 20); min samples per leaf (1–19); max features (sqrt, 0.3, 0.5)	10

Model	Hyperparameters (range)	Candidates
Hist gradient boosting	Learning rate (0.02–0.3, log); iterations (100–499); max leaf nodes (8–63); min samples per leaf (10–99); L2 (10^{-3} to 10, log)	20
Linear SVM	C (log-uniform 10^{-3} to 10)	8

Three further experiments used the traditional models. A learning curve refitted the best model on growing fractions of the training data. A feature-family ablation (ML-3) retrained gradient boosting on six feature subsets. A prediction-day sensitivity analysis (ML-4) rebuilt the entire cohort and feature set at days 0, 30, 60, 90 and 120.

G. Deep Learning

Networks were built in TensorFlow [22] with the Keras API and fed by `tf.data` pipelines that cache the data, reshuffle it each epoch, batch it and prefetch the next batch. Sequential models took the dense 91-column matrix; the Functional model took eight integer-coded categorical tensors plus one numeric tensor.

DL-1 is a small Sequential multilayer perceptron (MLP) with hidden layers of 64 and 32 ReLU units and no regularisation. DL-2 enlarges it to 256-128-64 units, and DL-3 adds dropout (0.3) [23], L2 weight decay (10^{-4}) and batch normalisation [24], isolating the effects of capacity and regularisation. DL-4 is a ten-trial random search over five layer configurations, dropout (0.1–0.5), L2 penalty (0 to 10^{-3}), batch normalisation, optimiser (Adam [25], RMSprop or Nesterov SGD), learning rate (3×10^{-4} to 3×10^{-3}) and batch size (128–512). DL-5 is a Functional-API network in which each categorical feature passes through its own learned embedding [15] of dimension $\min(8, \lceil(\text{levels} + 1)/2\rceil)$; the embeddings are concatenated with the numeric features and fed to the best DL-4 stack. All networks use a sigmoid output and binary cross-entropy, and train for up to 80 epochs with early stopping on validation ROC-AUC (best weights restored) and learning-rate halving when validation loss plateaus.

H. Evaluation Protocol

ROC-AUC is the headline metric because it measures ranking quality independently of any threshold, which suits a tutor working down a ranked list. PR-AUC, precision, recall and F1 for the at-risk class are reported alongside it, since precision–recall analysis is more informative about the positive class [26]. The best model from each paradigm was selected on validation ROC-AUC. A decision threshold was chosen on the validation set to maximise F1 and then applied unchanged to the test set. The difference in test ROC-AUC between the two finalists was assessed with a 1,000-sample bootstrap [27], and calibration was checked with reliability curves [28]. Errors were then analysed by original four-class outcome, by module and by demographic group, and permutation importance was computed on the full pipeline. Finally, a temporal robustness check retrained the best traditional model on the three earlier presentations and tested it on the last (2014J).

IV. Results

A. Traditional Models: Baselines and Tuning

All four traditional families clearly beat the dummy classifier (ROC-AUC 0.500), and their validation scores fell within a narrow band (Table III). The default random forest reached a training ROC-AUC of 1.000, meaning it had memorised the training data, while logistic regression's training and validation scores differed by only 0.004.

TABLE III. Validation ROC-AUC of traditional models before and after tuning

Model	Default	Tuned	Training (tuned)	Selected hyperparameters
Logistic regression	**0.8373**	0.8372	0.841	C = 0.075, no class weighting
Hist gradient boosting	0.8356	0.8358	0.862	learning rate 0.071, 406 iterations, 14 leaves, L2 = 8.57
Random forest	0.8307	0.8326	0.966	300 trees, depth 20, min leaf 7, sqrt features
Linear SVM	0.8137	0.8294	0.833	C = 0.0017
Dummy (class prior)	0.5000	–	–	–

Tuning changed validation ROC-AUC by less than 0.002 for three families. Only the linear SVM gained meaningfully (0.016), by choosing much stronger regularisation. Every family's selected hyperparameters favoured regularisation: gradient boosting chose small, heavily penalised trees, which cut its training score from 0.903 to 0.862 with no loss on validation. Across its 20 candidates, validation ROC-AUC stayed between 0.830 and 0.835, while training ROC-AUC rose from 0.865 to 0.925 as trees grew. Cross-validated and validation scores agreed within 0.002 for every family, so the validation set was representative.

The learning curve for logistic regression (Fig. 3) shows the cross-validation score rising from 0.76 with about 1,200 training enrolments to 0.835 with about 12,300. Over the same range the training score falls from 0.86 to 0.843, and the two curves almost meet. The model is therefore limited mainly by bias, meaning the information in its features, rather than by variance. More data would bring only a small further gain.

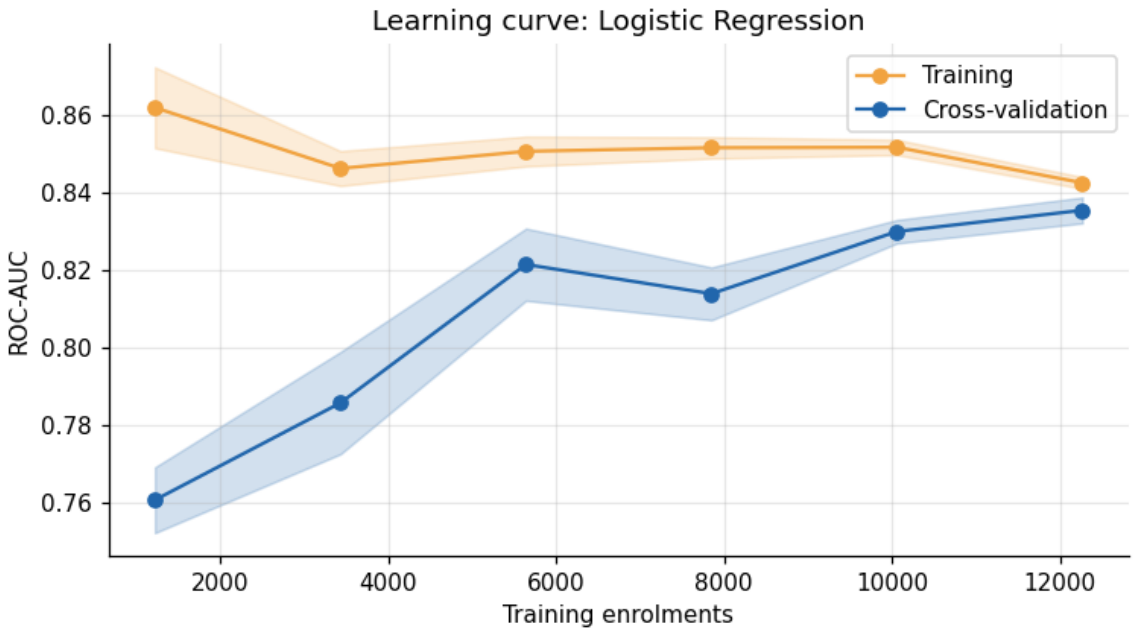


Fig. 3. Learning curve for logistic regression (three-fold grouped cross-validation; shading shows ±1 standard deviation).

B. Feature Ablation and Prediction Day

The ablation (Fig. 4) shows that behaviour is far more predictive than background. Background features alone, which is everything known on enrolment day, gave a validation ROC-AUC of 0.649. Assessment features alone gave 0.810, and VLE engagement features alone gave 0.765. The two behavioural families were complementary: adding VLE features to background and assessment raised ROC-AUC from 0.816 to 0.836 and recall at a threshold of 0.5 from 0.590 to 0.633.

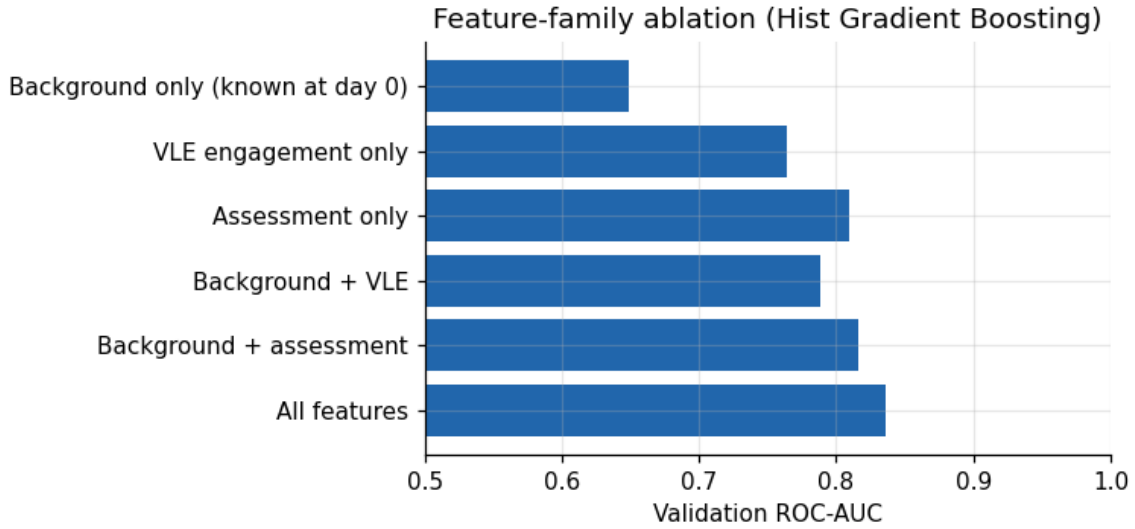


Fig. 4. Validation ROC-AUC of histogram gradient boosting trained on different feature families.

Rebuilding the cohort and features at different prediction days (Fig. 5) showed a steady trade-off between earliness and accuracy. ROC-AUC was 0.721 at day 0, 0.787 at day 30, 0.836 at day 60, 0.859 at day 90 and 0.894 at day 120. Meanwhile the cohort shrank from 29,262 to 24,775 enrolments as students left. The day-0 model outperformed the background-only ablation (0.721 against 0.649) because VLE activity before the module starts already carries signal.

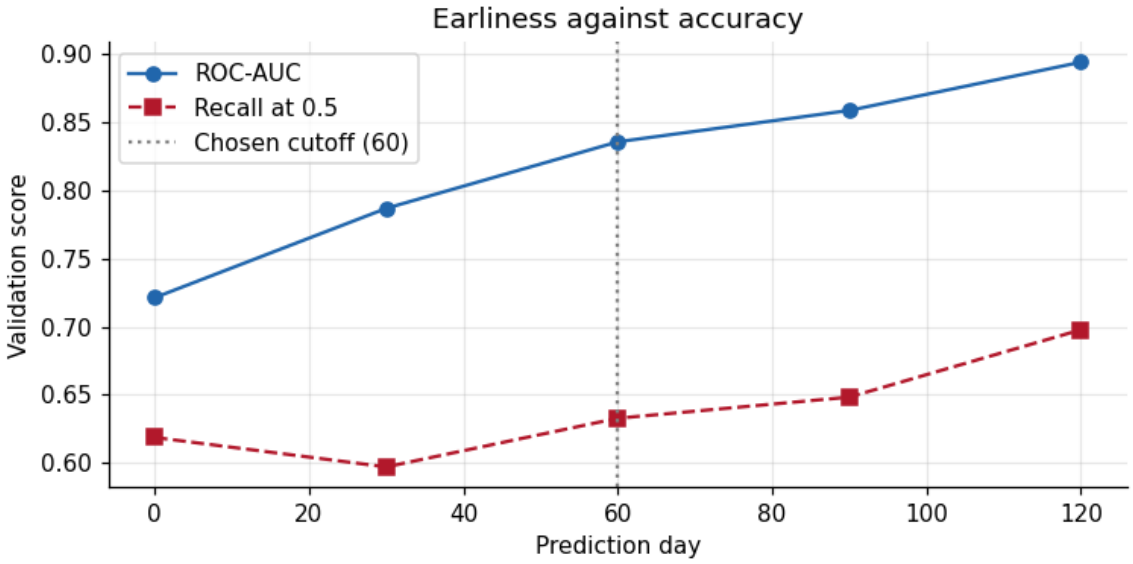


Fig. 5. Validation ROC-AUC and recall (threshold 0.5) for models rebuilt at prediction days 0–120.

C. Deep Learning Experiments

The capacity experiments produced three distinct training patterns (Fig. 6, Table IV). DL-1's validation loss plateaued by epoch 3 while its training loss kept falling. DL-2 overfitted in the classic way: validation loss reached its minimum (0.477) at epoch 3 and rose to 0.555 by epoch 13, while validation ROC-AUC fell from 0.841 to 0.813. Early stopping restored the epoch-3 weights, so its reported score remained competitive. DL-3 kept its curves close together and peaked later (epoch 10); its training loss initially sat above validation loss because dropout is active only during training.

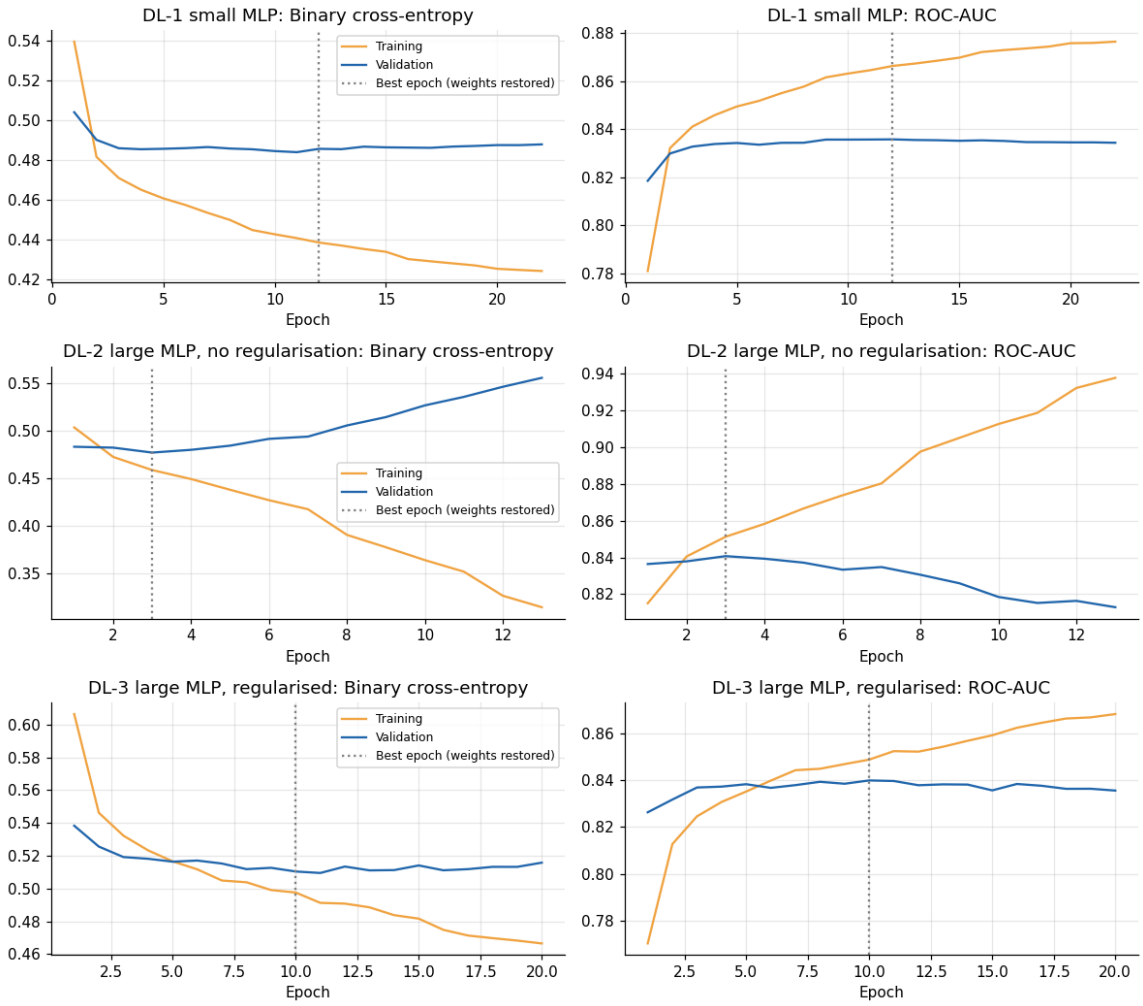


Fig. 6. Training (orange) and validation (blue) loss and ROC-AUC per epoch for DL-1 (top), DL-2 (middle) and DL-3 (bottom). Dotted lines mark the epoch whose weights were restored.

TABLE IV. Deep learning experiments (validation set)

Experiment	Architecture and settings	Parameters	Best epoch	ROC-AUC	PR-AUC
DL-1	64-32, no regularisation, Adam	8,001	12 of 22	0.8359	0.8207
DL-2	256-128-64, no regularisation, Adam	64,769	3 of 13	0.8408	0.8261
DL-3	256-128-64, dropout 0.3, L2 10^{-4}, batch norm	66,561	10 of 20	0.8400	0.8255
DL-4 best (trial 10)	256-128-64, dropout 0.3, RMSprop, lr 3×10^{-4}, batch 256	64,769	24 of 32	**0.8411**	0.8261

Experiment	Architecture and settings	Parameters	Best epoch	ROC-AUC	PR-AUC
DL-4 worst (trial 03)	128-64-32, dropout 0.4, batch norm, SGD, lr 10^{-3}	23,041	80 of 80	0.8272	0.8164
DL-5	Functional, 8 embeddings + best DL-4 stack	60,691	32 of 42	0.8402	0.8262

The random search gave validation ROC-AUC from 0.827 to 0.841, with the top six trials within 0.0013 of one another. The two weakest trials used SGD and were still improving at the 80-epoch limit. Trial 09 used the same optimiser and learning rate as trial 02 but half the batch size, and converged by epoch 44 to 0.841, so the weak trials were limited by their training budget rather than their architecture. The embedding model (DL-5) matched but did not beat the best one-hot MLP.

D. Final Comparison on the Test Set

On the test set (Table V), the best neural network (DL-4 trial 10) reached a ROC-AUC of 0.835 and the best traditional model, the default logistic regression, 0.831. Both scored about 0.007 below their validation results, the small drop expected after selecting the best of many candidates. The ROC and precision–recall curves of all models almost coincide (Fig. 7).

TABLE V. Test-set performance (3,971 enrolments; threshold 0.5 unless stated)

Model	ROC-AUC	PR-AUC	Accuracy	Precision	Recall	F1
MLP, DL-4 trial 10	**0.8346**	**0.8132**	0.768	0.770	0.637	0.697
MLP, DL-4 trial 10 (threshold 0.35)	0.8346	0.8132	0.730	0.644	**0.794**	**0.711**
Embedding network, DL-5	0.8345	0.8138	0.767	0.768	0.635	0.695
Logistic regression	0.8306	0.8081	0.765	0.771	0.625	0.691
Logistic regression (threshold 0.38)	0.8306	0.8081	0.748	0.689	0.727	0.707
Hist gradient boosting (tuned)	0.8276	0.8088	0.758	0.762	0.615	0.681
Random forest (tuned)	0.8271	0.8061	0.758	0.773	0.598	0.674
Linear SVM (tuned)	0.8218	0.7984	0.757	0.790	0.573	0.664
Dummy (class prior)	0.5000	0.4193	0.581	0.000	0.000	0.000

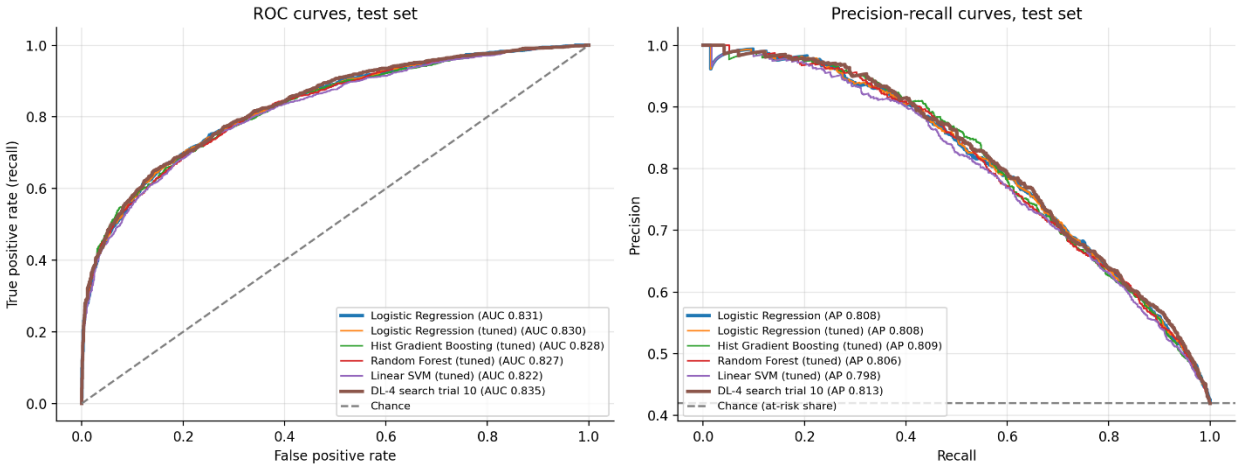


Fig. 7. ROC (left) and precision–recall (right) curves on the test set for the selected logistic regression, the tuned traditional models and the best neural network..

The 95% bootstrap interval for the difference in test ROC-AUC (logistic regression minus MLP) was [−0.0083, +0.0002], with a median of −0.0039. Because the interval includes zero, the difference is not statistically significant. Both models were well calibrated, with reliability curves within about 0.06 of the diagonal in every bin.

At the validation-tuned thresholds, the confusion matrices (Fig. 8) translate into concrete numbers of students. Of the 1,665 at-risk students in the test set, logistic regression flagged 1,210 (72.7%) and missed 455, and 546 of its 1,756 flags (31%) were on-track students. The MLP, at a lower threshold, flagged 79.4% of at-risk students but produced about 185 more false alarms. On the precision–recall curve, reaching 80% of at-risk students requires accepting a precision of about 0.64.

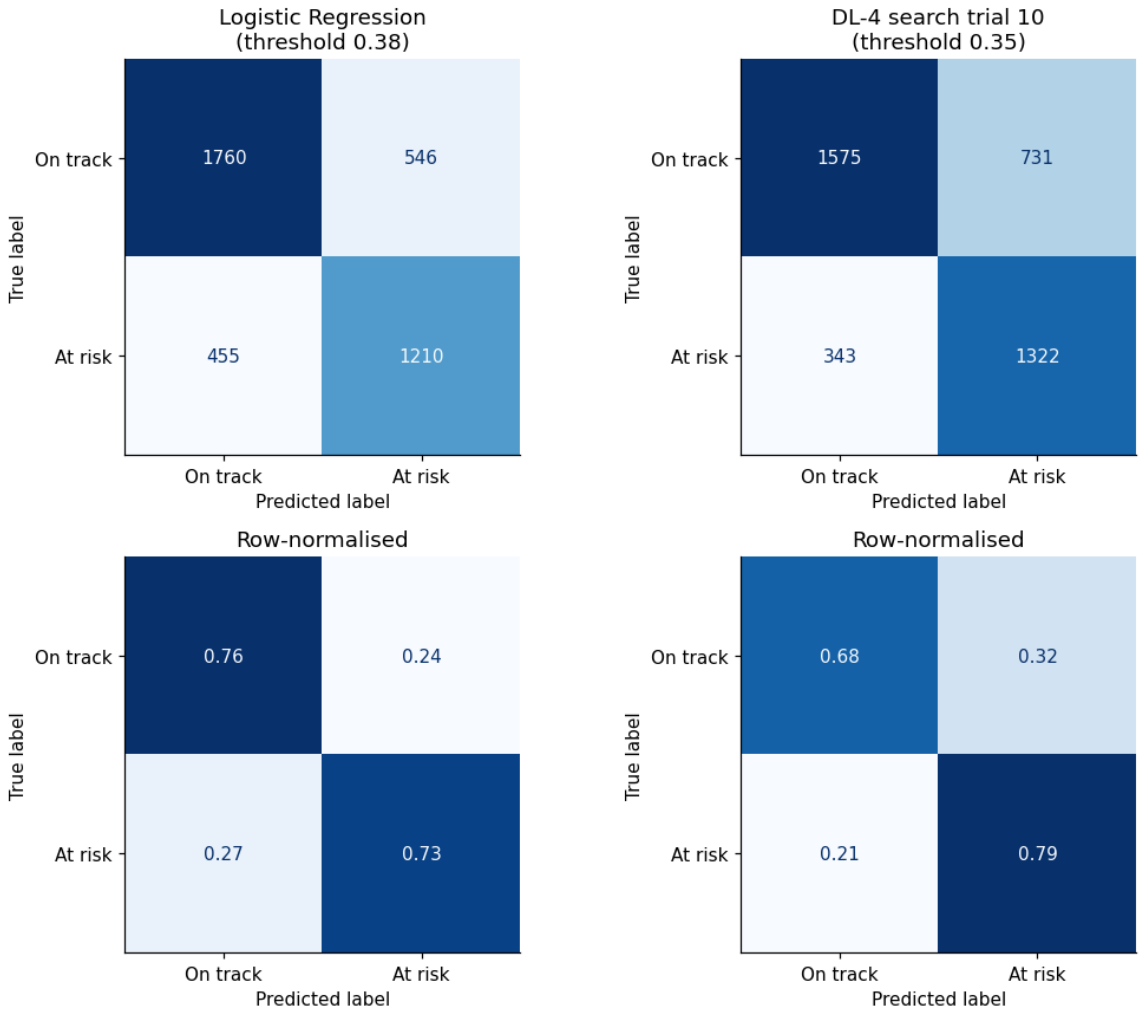


Fig. 8. Test-set confusion matrices for logistic regression (left) and the best MLP (right) at their validation-tuned thresholds; counts (top) and row-normalised rates (bottom).

E. Error Analysis

Breaking the predictions back into the four outcomes (Fig. 9) shows where errors concentrate. Logistic regression flagged 74.1% of students who later failed and 70.0% of those who withdrew, and it wrongly flagged 28.3% of Pass students but only 5.6% of Distinction students. Fail and Withdrawn students have a peak near 1.0 but a long tail that overlaps the Pass distribution. The two paradigms agreed on 90.8% of test predictions.

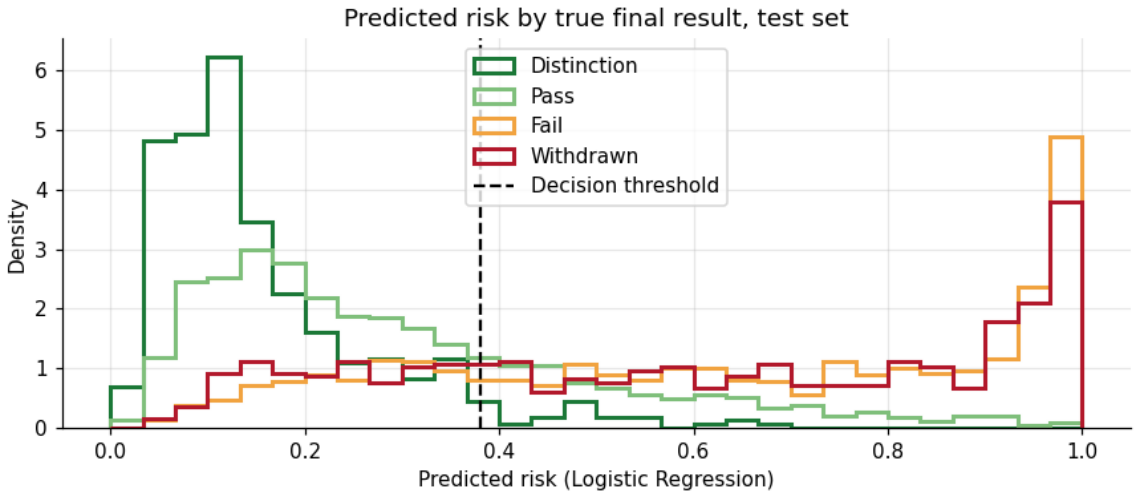


Fig. 9. Distribution of logistic-regression predicted risk on the test set, by true final result. The dashed line is the decision threshold.

The missed students resembled successful students. Their median profile was 473 clicks, 27 active days, a last login one day before the cutoff, a mean score of 81.7 and full submission. This is close to the true negatives (577 clicks, 32 days, score 85.0) and far from the true positives (190 clicks, 13 days, score 64.3).

Performance was consistent across six modules (ROC-AUC 0.82–0.85), but module GGG reached only 0.737, with recall of 0.545. Error rates also differed by group (Table VI). The false-positive rate rose with deprivation to 2.5 times that of the least deprived band, and it was higher for students with a declared disability despite identical ROC-AUC. Recall was lower for women and lowest for students whose deprivation band was missing.

TABLE VI. Test-set error rates by group, logistic regression at threshold 0.38

Group	n	At-risk rate	Recall	False-positive rate	ROC-AUC
IMD 0–10 (most deprived)	400	0.528	0.806	0.407	0.829
IMD 90–100 (least deprived)	326	0.316	0.718	0.161	0.868
IMD missing	174	0.322	0.536	0.127	0.787
Declared disability	348	0.514	0.810	0.367	0.830
No declared disability	3,623	0.410	0.717	0.226	0.830
Female	1,788	0.410	0.673	0.250	0.801
Male	2,183	0.427	0.769	0.225	0.853

Permutation importance (Fig. 10) ranked the weighted assessment score (an AUC drop of 0.046) and the module (0.045) highest. These were followed by the number of distinct VLE resources used (0.024), late

submissions (0.015) and days since last login (0.014). Highest education (0.008) and IMD band (0.005) were the only demographic features in the top 20.

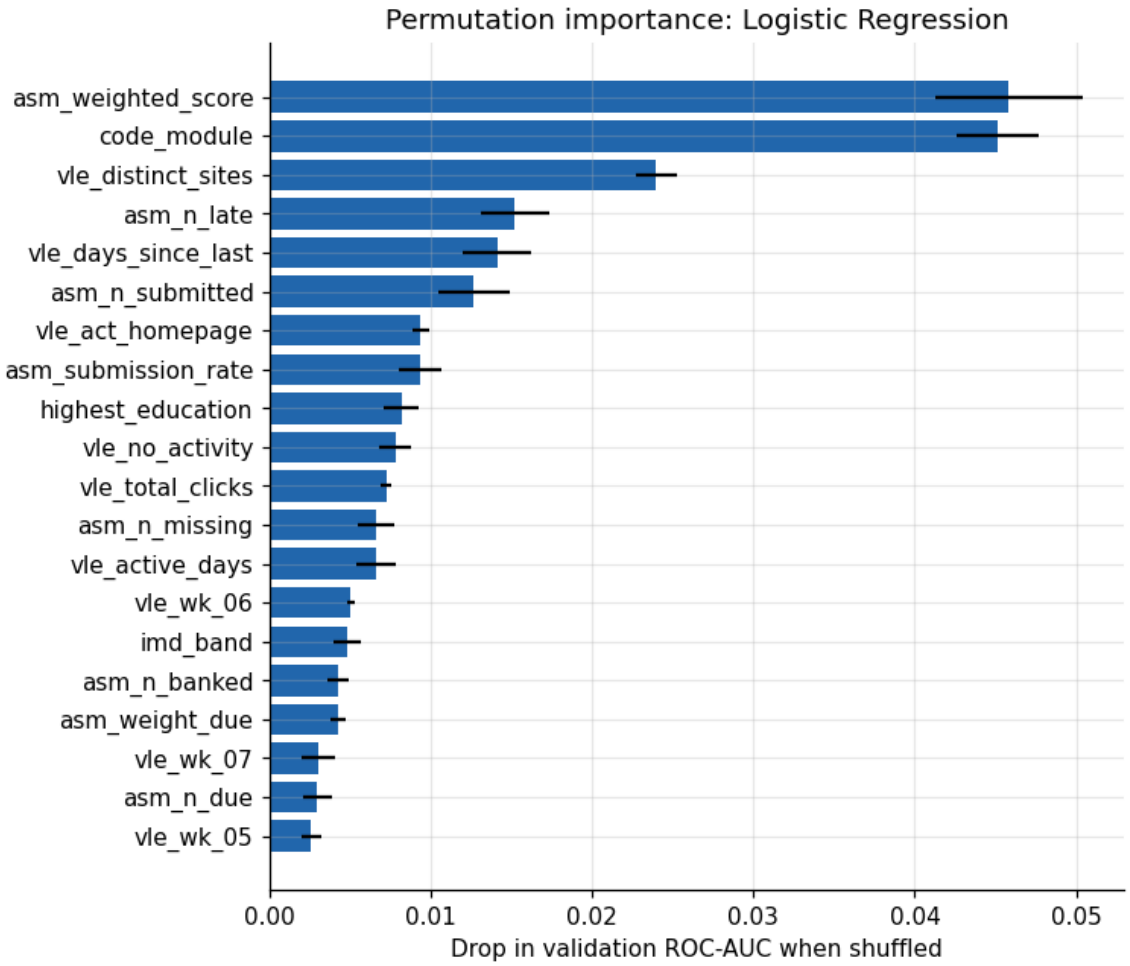


Fig. 10. Permutation importance of the 20 most important raw features for logistic regression (drop in validation ROC-AUC; bars show ±1 standard deviation over five repeats).

Finally, the temporal robustness check gave a ROC-AUC of 0.815 and a PR-AUC of 0.770 on the 2014J presentation (8,816 enrolments), compared with 0.831 and 0.808 on the random split. Recall at a threshold of 0.5 fell from 0.625 to 0.550.

V. Discussion

A. Traditional versus Deep Learning

The central finding is that the two paradigms performed the same. All 22 trained models, eight traditional and fourteen neural, reached validation ROC-AUC between 0.814 and 0.841. Every tuned model scored 0.822–0.835 on the test set, and the bootstrap interval for the finalists' 0.004 difference included zero. This agrees with tabular-data benchmarks [6], [7] and qualifies OULAD studies that report deep models ahead of baselines [11], [12]: with a strict cutoff, a student-disjoint split and matched tuning, the gap disappears.

The learning curves explain why. Logistic regression's training and cross-validation curves nearly converged (Fig. 3), and every flexible model turned extra capacity into a higher training score without any validation gain (Fig. 6). The models are limited by the information in the features, not by their ability to fit them. The error analysis confirms this: the missed students looked, on every behavioural measure, like students who passed, so their outcome was not visible at day 60.

With equal accuracy, practical grounds favour logistic regression. It trains in about one second rather than 25, needs no separate preprocessing artefact, is well calibrated, and has coefficients that can be explained to tutors and to the students being scored. The MLP's higher recall at its chosen threshold comes from the threshold rather than a better ranking, and logistic regression could match it by lowering its own.

B. What Drives Risk, and When

The ablation, correlation and permutation analyses tell a consistent story. Early assessment results are the strongest single signal, and engagement adds complementary information. Within engagement, consistency and recency (weeks active, distinct resources, days since last login) matter more than raw click volume. For practice, this means a tutor's most useful early indicators are a missing or low-scoring first assignment and a student who has gone quiet, not simply a student who clicks less than others. The prominence of the module itself in the importance ranking echoes Gašević et al. [8]: base rates and the meaning of behaviour differ between courses, so a single pooled model must at least know which course it is scoring. GGG's weak performance suggests that some modules may need their own models or features.

The prediction-day experiment frames the operational trade-off. Moving from day 0 to day 60 raised ROC-AUC from 0.721 to 0.836, but 61.9% of all withdrawals had already happened by day 60 (Section III-B). A single day-60 model therefore cannot help most early withdrawers. A better design would be a sequence of models: a coarse day-0 or day-30 screen to prompt light-touch welcome contact, followed by sharper day-60 and day-90 models to direct intensive support. This mirrors the weekly prediction cycle of OU Analyse [9].

C. Fairness and Responsible Use

The subgroup analysis shows a pattern documented elsewhere in education [16], [17]. The model ranks students equally well within each group but applies a higher baseline risk to disadvantaged groups, so their on-track members are wrongly flagged more often. The false-positive rate in the most deprived band (0.407) is 2.5 times that of the least deprived (0.161). This partly reflects real differences in outcome rates (52.8% against 31.6% at risk), and demographic features carry little weight in the model, so the disparity is driven mainly by correlated behaviour and context. Removing demographic features would therefore not remove it.

Whether this is acceptable depends on what the flag triggers. If it triggers a supportive email or a tutor phone call, an unnecessary contact for a deprived student is a small cost, and arguably a benefit. If it fed into anything restrictive, such as limiting course choices or flagging students to funders, the same disparity would entrench disadvantage. A responsible deployment would use the score only for support, monitor recall and false-positive rates by group each presentation, and consider group-aware thresholds where equal error rates are required. The lower recall for women and for students with missing

deprivation data deserves further investigation, because it means those at-risk students are more likely to be overlooked.

D. Limitations

Several limitations constrain these conclusions. The data cover one institution, seven modules and two years, so the behaviour patterns may not transfer to other universities or current platforms; even one presentation ahead, ROC-AUC fell by 0.016. The cohort excludes students who left before day 60, and the binary label merges academic failure with withdrawal for personal or financial reasons, which call for different support. Clicks are only a proxy for learning, and scores were assumed to be available on the submission date, ignoring marking delays. Finally, the searches were modest and used a single seed, so differences below about 0.002 ROC-AUC are noise, and the enrolment-level bootstrap slightly understates uncertainty.

VI. Conclusion

This study asked whether students who will fail or withdraw from a distance-learning module can be identified 60 days after it starts. Using a leakage-free, student-disjoint version of OULAD, the answer is a qualified yes. The best models rank a randomly chosen at-risk student above a randomly chosen on-track student about 83% of the time (ROC-AUC 0.831–0.835). At a recall-oriented threshold, they reach 73–79% of at-risk students, with roughly two in three flags correct.

The comparison between paradigms produced a clear and practically useful result. A tuned three-layer MLP and an untuned logistic regression were statistically indistinguishable, and neither regularisation, random search nor categorical embeddings moved the neural networks beyond the ceiling the traditional models reached. The learning curves and error analysis locate that ceiling in the data. Students whose difficulties had not yet surfaced by day 60 cannot be detected from day-60 features. For this problem, effort is better spent on what is measured and when than on model complexity. Assessment and engagement features raised ROC-AUC from 0.649 to 0.836, while the choice of algorithm changed it by less than 0.01.

The analysis also identified risks that a deployment must manage. Error rates were unequal across socio-economic and disability groups, performance varied by module, and accuracy dropped slightly on a later cohort. These findings argue for using the score only to offer support, monitoring error rates by group, and retraining each presentation.

Future work could extend this study in four directions. Sequence models such as LSTMs [29] or one-dimensional convolutions over the daily click series could exploit the order of events that weekly aggregates discard. A multi-class or time-to-event formulation could separate withdrawal from failure, which need different interventions. Fairness-aware thresholding could be evaluated against the subgroup error rates reported here. Most importantly, a controlled trial would be needed to show that acting on these predictions actually improves retention, because predicting risk is not the same as reducing it.

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